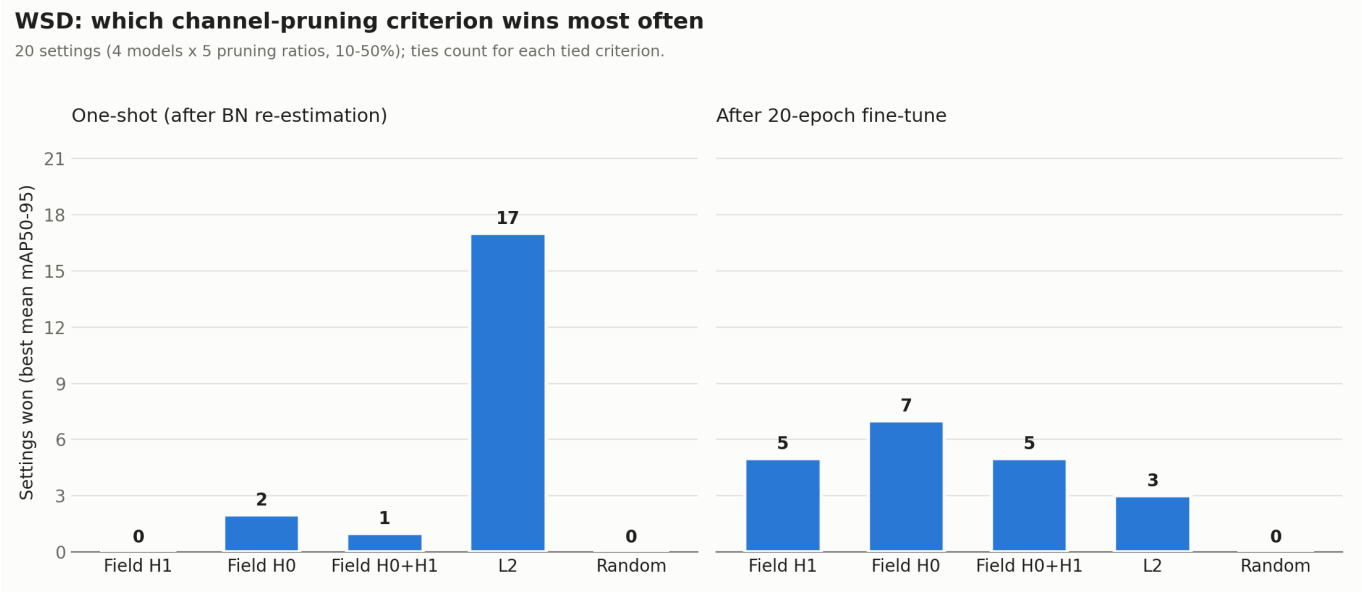
Pruning YOLO26 and YOLOv5 on WSD, Viewsphere and COCO: results to date

Blake J. Gisclair

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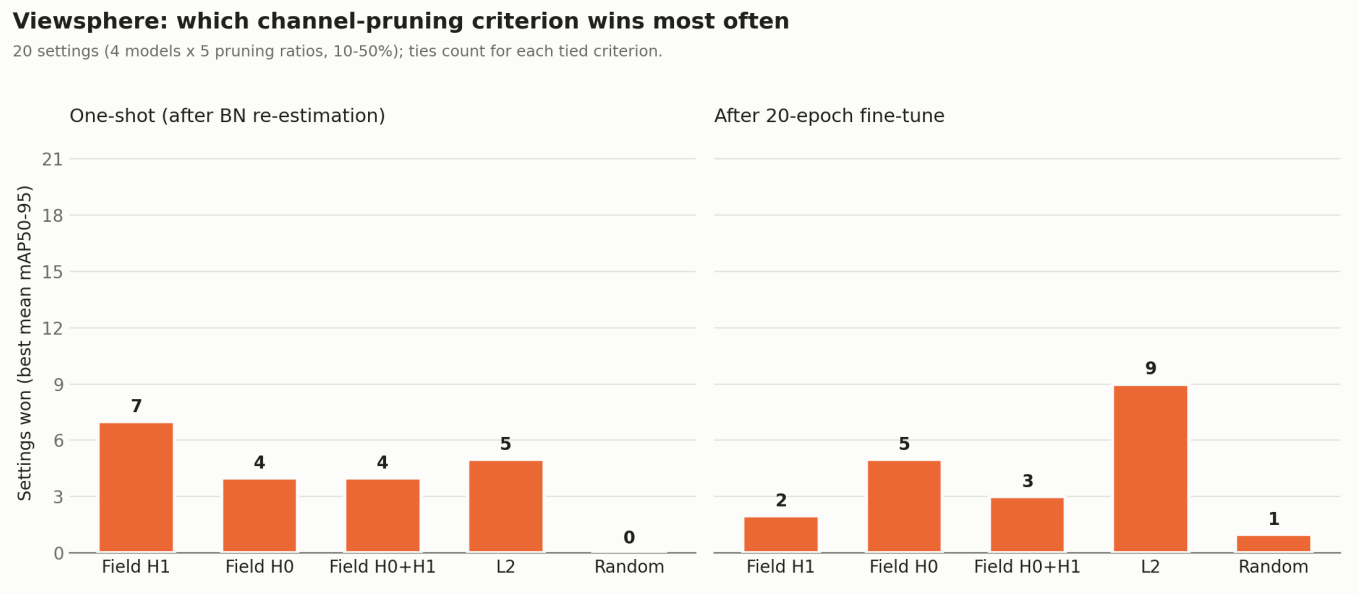
Summary. On COCO, PEEK field H1 beats L2 and L2 beats random at every pruning ratio, before and after fine-tuning. On WSD and Viewsphere, L2 is as good as or better than field H1 right after pruning (both clearly beat random), and after a 20-epoch fine-tune all three end within 0.01–0.04 mAP50-95 of each other. Scoring channels by H0 or by H0+H1 instead of H1 changes little: on COCO all three persistence scores beat L2 after fine-tuning, and elsewhere they trade small leads. For block replacement the score matters more, and the best ranking differs by dataset (H1 only on WSD, the H0/H1 blend or H0 only on COCO). Channel pruning cuts up to 28% of GFLOPs but only 1–4% of GPU latency. Repeating everything on YOLOv5 (Section 4) keeps the broad picture: persistence scores lead on COCO, but by less and with L2 no better than random; criteria converge after fine-tuning on WSD and Viewsphere; and the best block ranking per dataset is the same as on YOLO26. On Viewsphere v2 (17 satellites, Section 5), no channel criterion beats the runner-up by more than the seed noise after fine-tuning, and the raw head ranking is again the block ranking that beats random choice. Figures 1–5 count how often each channel criterion comes out best: one figure per dataset, then all four together. A setting = model (YOLO26n/s, YOLOv5n/s) × pruning ratio (10–50%), 20 per dataset; the criterion with the highest mean mAP50-95 over seeds wins it (ties count for each). A win is a highest mean, not necessarily a lead beyond the seed noise (the † in the tables).

Figure 1: WSD: which channel-pruning criterion wins most often



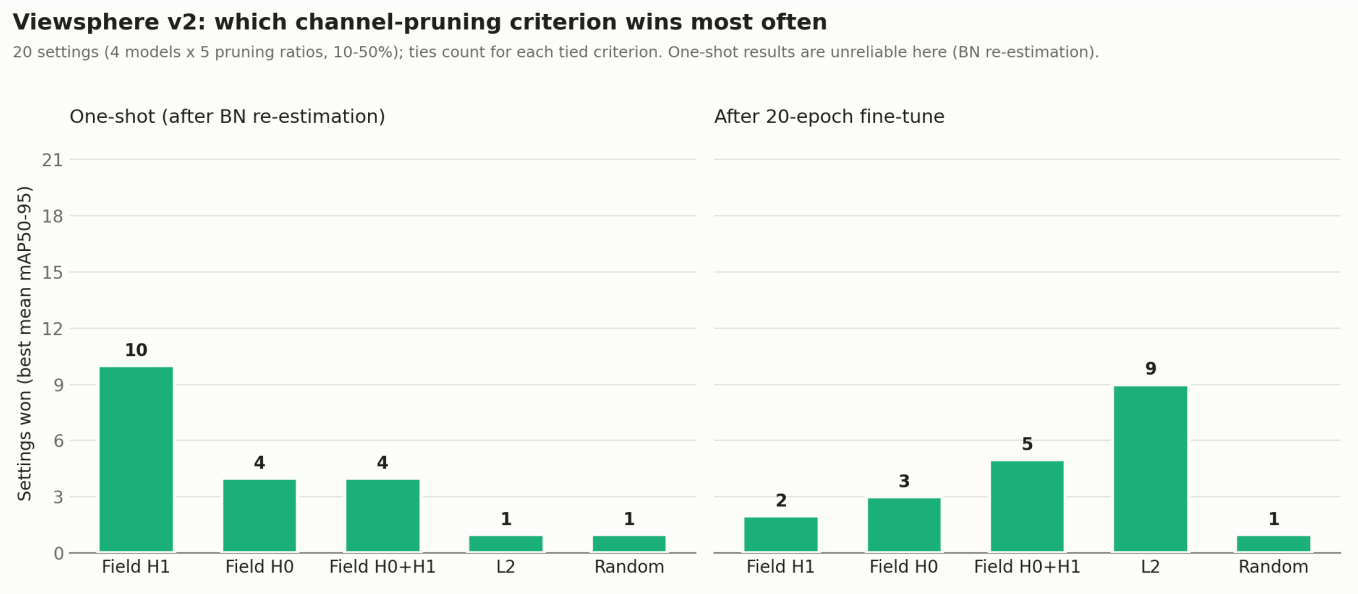
One-shot, L2 wins 17 of 20. After fine-tuning the persistence scores win 17 (field H0 7, field H1 and H0+H1 5 each), L2 3 and random none.

Figure 2: Viewsphere: which channel-pruning criterion wins most often



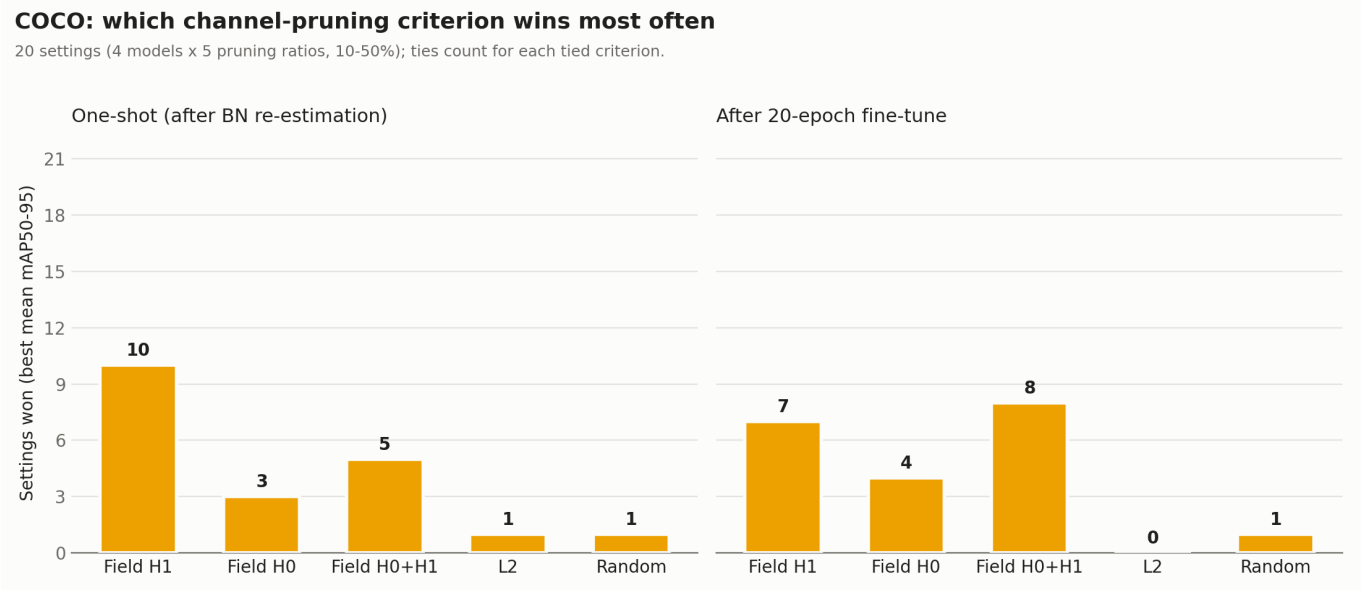
One-shot, field H1 wins most often (7 of 20); after fine-tuning, L2 does (9 of 20).

Figure 3: Viewsphere v2: which channel-pruning criterion wins most often



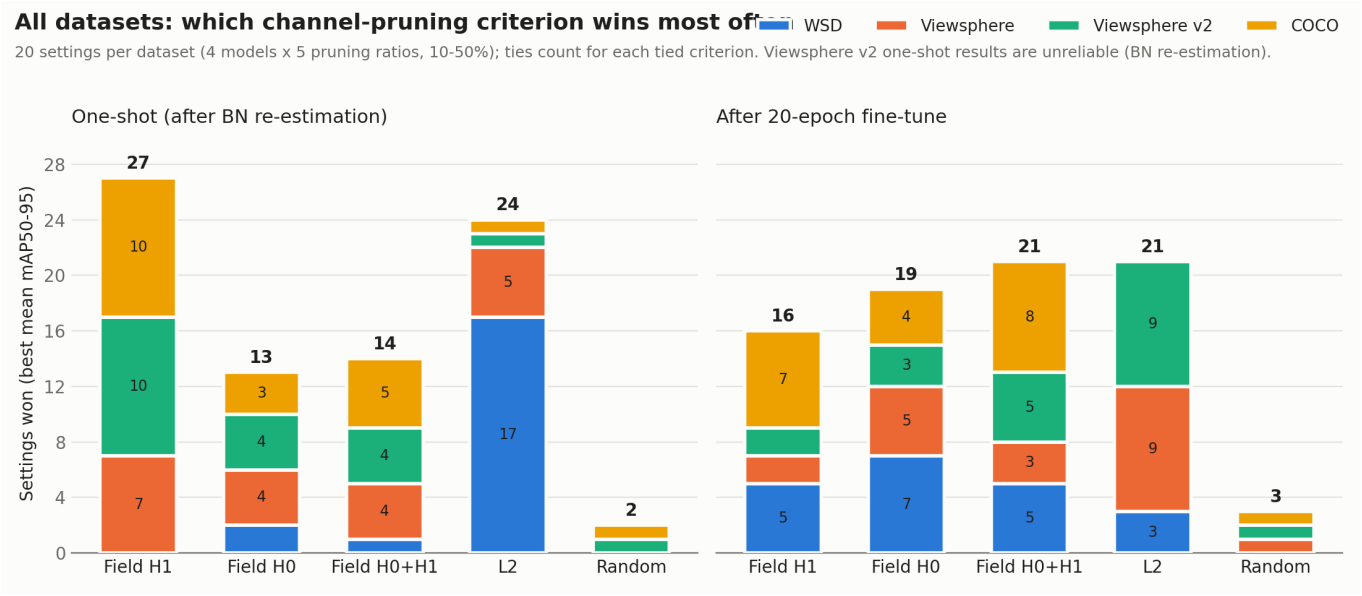
After fine-tuning, L2 wins most often (9 of 20), as on Viewsphere. The one-shot counts (field H1 10 of 20) are unreliable: BN re-estimation collapses some models on v2 (Section 5).

Figure 4: COCO: which channel-pruning criterion wins most often



One-shot, field H1 wins half the settings (10 of 20). After fine-tuning a persistence score wins 19 of 20 (field H0+H1 8, field H1 7, field H0 4); L2 wins none.

Figure 5: All datasets: which channel-pruning criterion wins most often



Figures 1–5 stacked. One-shot, field H1 (27 of 80) and L2 (24, 17 of them on WSD) win most often. After fine-tuning the four scored criteria win about equally often (16–21 each), and random almost never (3).

Win counts say which criterion is best, not by how much. Figure 6 shows the margin: how far each persistence score is from L2 at each pruning ratio.

Figure 6: How close the persistence scores are to L2



Criterion mAP50-95 minus L2's at the same dataset, model and pruning ratio, averaged over YOLO26n/s and YOLOv5n/s; 0 = L2, above = better. The three persistence scores move together, mostly within 0.01 of each other. One-shot, they trail L2 on WSD (by up to 0.07 at 30–40%) and lead it elsewhere (COCO: about +0.04 at 10–20%, shrinking to 0 by 50%). After fine-tuning they are level with L2 on WSD (within 0.004), slightly behind on Viewsphere (up to −0.015), scattered around it on Viewsphere v2 (+0.014 to −0.030), and ahead on COCO by a margin that grows with pruning (+0.005 at 10% to about +0.04 at 50%). The averages hide model-to-model variation, and many of the small gaps are within seed noise; Viewsphere v2 one-shot results are unreliable (Section 5).

1 Datasets

Table 1: Datasets and the starting models that are pruned

Dataset	Content	Train / val images	Starting model
WSD v71	satellite components: antenna, body, solar, thruster	1,005 / 186	YOLO26n/s trained from scratch
Viewsphere	4 rendered satellites, relabelled to WSD's 4 classes	925 / 160	the WSD models fine-tuned on it
Viewsphere v2	17 rendered satellites, same classes and split (Section 5)	4,079 / 680	the WSD models fine-tuned on it
COCO 2017	80 everyday object classes	10k subset / 5,000	Ultralytics pretrained YOLO26n/s

Viewsphere renders Clementine, ICESatB, Jason-1 and Sentinel-6 from 18 viewing angles, 4 distances and 5 Sun phases; it is split by viewing angle, and frames with less than 10% of the satellite lit are dropped from train. Training on it from scratch was unstable, hence the WSD starting point. On **COCO**, pruned models are fine-tuned on a fixed 10k-image subset of train2017, so the unpruned model given the same fine-tune is the reference.

2 Methods

Channel pruning. In every convolution from layer 6 on (the stem, the detection head and attention layers are left alone), the lowest-scoring 10–50% of channels are removed physically. All criteria remove the same number of channels per layer, so they give equally sized models; at 50% that is −39% parameters and −28% GFLOPs. The criteria compared:

- **PEEK field H1** — builds the layer's PEEK map (pseudo-entropy of the activations across channels at each pixel), measures its loops (H1 persistent homology), and removes first the channels whose absence changes that topology least. Scored on 128 training images.

- **PEEK field H0** and **field H0+H1** — the same, but measuring the map's connected components (H0) or the total persistence of both (H0+H1).
- **L2** — removes the filters with the smallest weight magnitude.
- **Random** — removes a random set of channels; the baseline to beat.

Block replacement. Whole C3k2 blocks are swapped for a 1×1 convolution, $k = 1, 2, 3$ blocks at a time, ranked by the persistence of each block's PEEK map: *head* ranks the four head blocks on raw scores; *grid* and *null* rank all eight C3k2 blocks with the scores normalised for map size. Each ranking is scored three ways: *C*, the 50/50 blend of normalised H0 (secondary-peak prominence) and H1 (pocket persistence) used so far, *H0* only, or *H1* only. A cut set picked by several rankings is trained once. The baseline is 23 random block orders.

Measurements (mAP50-95 on the validation split):

- **One-shot** — right after pruning, with only the BatchNorm statistics re-estimated; 10 seeds (random: 10 different masks).
- **After a 20-epoch fine-tune** — AdamW, batch 16; lr 1.25×10^{-3} (WSD, Viewsphere) or 1.25×10^{-4} (COCO); 5 seeds. Block cuts get the same fine-tune, 3 seeds.
- **Recovery** — separate fine-tunes stopped after 250 to 5,000 batches; 3 seeds on WSD and Viewsphere, 1 on COCO (which also has runs on all of train2017).

A difference is treated as real when it exceeds two standard errors across seeds (roughly a 95% test); otherwise it is within seed noise.

3 Results

Each dataset has a channel-pruning table; recovery (Tables 5–7) and block replacement (Tables 8 and 9) are collected for all three at the end of this section.

3.1 WSD

Table 2: WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.352			0.360	
	Field H1	0.289	0.098	0.017	0.350	0.338	**0.342**
	Field H0	0.292	0.106	0.006	0.353	0.341	0.338
	Field H0+H1	**0.297**†	0.108	0.004	**0.354**	0.341	0.333
	L2	0.289	**0.167**†	**0.064**†	0.353	**0.344**	0.340
	Random	0.243	0.054	0.005	0.350	0.333	0.329
YOLO26s	Unpruned		0.380			0.381	
	Field H1	0.334	0.208	0.035	0.376	**0.371**	**0.365**
	Field H0	**0.347**	0.151	0.018	**0.382**	0.368	0.359
	Field H0+H1	0.344	0.160	0.024	0.378	0.368	0.363
	L2	0.342	**0.239**†	**0.087**†	0.380	0.363	0.358
	Random	0.313	0.127	0.006	0.372	0.364	0.358

mAP50-95 on the validation split, mean over seeds (one-shot: 10, fine-tuned: 5). **Bold** = best criterion; † = its lead over the runner-up holds up across seeds.

- **One-shot:** L2 is best from 20% on, by 0.02–0.07; field H1 is second but still well above random (+0.01 to +0.08 at every ratio).
- **After fine-tuning:** the criteria end within 0.013 of each other, and field H1 and L2 within about 0.02 of the unpruned model even at 50%. The few leads that hold up (L2 at 30% on n, field H1 at 30–50% on s) are under 0.01.

- **H0 and H0+H1:** one-shot, they edge out field H1 at 10–30% on n and at 10% on s, while field H1 is better at 50% (and at 30% on s); after fine-tuning all three persistence scores are within 0.009 of each other with no consistent winner.
- **Recovery:** fast; most models are within 0.02–0.03 of unpruned after 1,000 batches (16 epochs). At 50% a real gap of 0.013–0.016 remains at 5,000 batches, and the three criteria are within noise of each other by then.
- **Blocks:** with the blended score (C), the size-normalised grid ranking gives the best cut in 5 of 6 cases (on n at $k = 2$, 0.365 — as good as the uncut model). Scored on H1 alone, the null ranking beats random choice at every k on both models (+0.011 to +0.043); the raw head ranking is mostly below the random median whatever the score.

3.2 Viewsphere

Table 3: Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.526			0.566	
	Field H1	0.410	0.144	0.043	0.551	0.517	0.456
	Field H0	0.415	0.127	0.033	**0.557**	0.519	0.466
	Field H0+H1	0.421	**0.168**	**0.048**	0.554	0.516	0.470
	L2	**0.446**	0.161	0.017	0.554	**0.523**	**0.472**
	Random	0.351	0.056	0.008	0.552	0.512	0.434
YOLO26s	Unpruned		0.570			0.577	
	Field H1	**0.517**	0.273	**0.052**	0.570	0.554	0.516
	Field H0	0.487	0.210	0.046	**0.576**	**0.562**	0.501
	Field H0+H1	0.498	0.236	0.036	0.573	0.550	**0.530**
	L2	0.516	**0.296**†	0.022	0.556	0.548	0.527
	Random	0.465	0.188	0.015	0.550	0.539	0.505

mAP50-95 on the validation split, mean over seeds. Bold and † as in Table 2.

- **One-shot:** field H1 beats random everywhere (+0.035 to +0.09). L2 leads or ties at 10–30%, field H1 leads at 50% (both models). Viewsphere runs are noisier than WSD's (seed sd 0.01–0.04).
- **After fine-tuning:** all within 0.04; L2 is slightly ahead at most ratios (by 0.006–0.016 where it holds up) and beats random by 0.012–0.038 at 40–50%.
- **H0 and H0+H1:** one-shot, field H1 is the best of the three on s; after fine-tuning, field H0+H1 beats field H1 at 50% on both models (0.470 vs 0.456 and 0.530 vs 0.516), roughly level with L2.
- **Recovery:** YOLO26s is back to the unpruned level by 5,000 batches even at 50%; YOLO26n stays 0.06 behind at 50%. L2 recovers slightly faster than field H1 here.
- **Blocks:** only the raw head ranking (the same cut for all three scores) beats the random median (by 0.012–0.021 at $k = 2, 3$). Scoring on H1 alone avoids the null ranking's loss at $k = 1$ (-0.030 and -0.033 become +0.005 and -0.003), but no all-block ranking beats random at $k \geq 2$.

3.3 COCO

Table 4: COCO: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.389			0.370	
	Field H1	0.195	**0.021**†	**0.002**†	0.341	**0.272**	**0.187**†
	Field H0	**0.218**†	0.015	0.001	0.341	0.271	0.175
	Field H0+H1	0.214	0.019	0.001	**0.342**	0.272	0.178
	L2	0.192	0.009	0.000	0.336	0.246	0.126
	Random	0.150	0.003	0.000	0.333	0.237	0.104
YOLO26s	Unpruned		0.460			0.424	
	Field H1	**0.338**	**0.060**†	**0.004**†	**0.406**	0.353	0.276
	Field H0	0.326	0.049	0.002	0.405	0.355	0.274
	Field H0+H1	0.336	0.055	0.004	0.405	**0.356**	**0.277**
	L2	0.279	0.019	0.001	0.400	0.335	0.212
	Random	0.244	0.006	0.000	0.401	0.334	0.203

mAP50-95 on val2017, mean over seeds. Bold and † as in Table 2.

- **One-shot:** far more fragile than WSD (near zero from 30–40%), but the order is clean: field H1 > L2 > random at every ratio, and every lead holds up.
- **After fine-tuning:** the same order survives and the gap grows with pruning; at 50%, field H1 0.187 vs L2 0.126 vs random 0.104 (n) and 0.276 vs 0.212 vs 0.203 (s).
- **H0 and H0+H1:** also ahead of L2 at every ratio after fine-tuning, and within 0.012 of field H1. One-shot, field H1 is the best of the three from 30% on (field H0 at 10% on n); after fine-tuning, field H1's lead at 50% on n (0.187 vs 0.175 and 0.178) holds up.
- **Recovery:** the 10k subset caps it, and field H1 recovers fastest. On all of train2017 (7 epochs), 10% pruning reaches 96–97% of the unpruned model and 50% reaches 73–83%, with field H1 still about 0.02 ahead of L2.
- **Blocks:** the null ranking is best on YOLO26n (at $k = 3$, 0.265 vs a random median of 0.144), grid on YOLO26s at $k \leq 2$; scored on H0 alone, null beats random at every k on YOLO26s (+0.033 to +0.063), while H1 alone is worse than random at $k = 1$. One cut block already costs 0.025–0.06 mAP.

3.4 Recovery, all datasets

Which criterion recovers first? On WSD and Viewsphere the differences are small and mostly early: at 250–1,000 batches, field H1 and L2 lead random by 0.01–0.05 at 50% on YOLO26n, but by 5,000 batches the three are within seed noise on WSD, and L2 is slightly ahead on Viewsphere. On COCO, field H1 leads at every budget at 30–50% pruning, most of all early (50%, 1,000 batches, YOLO26n: 0.077 vs 0.039 vs 0.022), and still after 7 epochs on all of train2017 (Table 6). Light pruning (10%) reaches 95% of the unpruned model within 250–500 batches on WSD and Viewsphere, but on COCO only after 3–7 epochs on the full training set; 30–50% never gets there on COCO (Table 7). Field H0 and field H0+H1 recover like field H1: on COCO all three stay well ahead of L2 at 50% (1,000 batches, YOLO26n: 0.072 and 0.076 vs 0.077 for field H1), and on WSD and Viewsphere most gaps to field H1 and L2 are within seed noise; those that hold up are small and point both ways on WSD ($\leq$0.013), while on Viewsphere L2 is often ahead of them by 0.01–0.04.

Table 5: Recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.353	0.357	0.358	0.363	0.363	0.361	0.373	0.381	0.386	0.390
	Field H1 10%	0.333	0.347	0.352	0.354	0.351	**0.359**	0.370	0.376	**0.384**	0.387
	Field H0 10%	0.339	0.350	0.352	0.355	**0.362**	0.350	**0.372**	0.375	0.381	0.385
	Field H0+H1 10%	0.341	**0.351**	0.353	**0.357**	0.356	**0.359**	0.369	**0.377**	0.382	0.388
	L2 10%	**0.342**	0.347	**0.354**	0.355	0.357	**0.359**	0.371	0.374	0.379	0.388
	Random 10%	0.335	0.345	0.347	0.356	0.355	0.351	0.369	0.373	0.381	**0.390**
	Field H1 30%	**0.321**	0.330	0.339	0.343	0.348	**0.346**	0.356	**0.368**	0.372	**0.383**
	Field H0 30%	0.314	0.332	0.337	**0.350**	**0.353**	0.339	0.353	0.356	**0.374**	0.379
	Field H0+H1 30%	0.311	0.326	0.340	0.347	0.347	**0.346**	**0.358**	0.361	0.371	0.381
	L2 30%	0.319	**0.337**	**0.341**	0.345	0.346	0.341	0.357	0.364	0.370	0.379
	Random 30%	0.303	0.320	0.329	0.341	0.342	0.342	0.354	0.359	0.370	0.380
	Field H1 50%	0.282	0.316	**0.337**	0.342	**0.347**	0.332	**0.356**	**0.365**	0.364	0.377
	Field H0 50%	**0.285**	0.315	0.335	**0.344**	0.346	0.332	0.351	0.358	0.360	0.369
	Field H0+H1 50%	0.284	0.312	0.329	0.339	0.344	**0.334**	0.347	0.359	0.364	0.373
	L2 50%	**0.285**	**0.319**	0.336	0.341	0.343	0.331	0.343	0.355	0.364	0.370
	Random 50%	0.257	0.307	0.327	0.335	0.341	0.322	0.346	0.361	**0.367**	**0.381**
Viewsphere	Unpruned	0.541	0.555	0.562	0.570	0.573	0.564	0.571	0.576	0.580	0.557
	Field H1 10%	0.504	0.531	0.547	0.558	0.550	0.542	0.560	**0.574**	0.585	0.587
	Field H0 10%	**0.529**	**0.545**	0.549	0.565	0.569	0.544	0.560	0.572	0.577	0.537
	Field H0+H1 10%	0.528	0.541	**0.550**	0.563	0.564	0.530	0.561	0.568	0.575	0.571
	L2 10%	0.523	0.544	0.549	0.565	**0.575**	**0.554**	**0.566**	0.573	**0.595**	**0.596**
	Random 10%	0.495	0.534	0.546	**0.568**	0.571	0.531	0.552	**0.574**	0.563	0.584
	Field H1 30%	0.414	0.485	0.511	0.532	0.546	0.491	0.534	0.553	0.570	0.580
	Field H0 30%	0.432	0.497	0.515	0.534	0.543	0.505	**0.546**	**0.560**	0.566	**0.585**
	Field H0+H1 30%	0.426	0.489	0.503	0.536	0.544	**0.506**	0.542	0.553	0.569	0.578
	L2 30%	**0.445**	**0.501**	**0.519**	**0.537**	**0.554**	0.491	0.530	0.558	0.564	0.542
	Random 30%	0.397	0.466	0.509	**0.537**	0.553	0.490	0.516	0.538	**0.576**	0.583
	Field H1 50%	0.292	0.408	0.457	0.482	0.512	0.401	0.480	0.520	0.539	0.549
	Field H0 50%	**0.321**	0.418	0.465	0.487	0.504	0.394	0.467	0.522	0.548	**0.572**
	Field H0+H1 50%	0.296	**0.421**	0.452	0.494	0.510	0.411	0.492	0.530	**0.552**	0.571
	L2 50%	0.310	**0.421**	**0.467**	**0.503**	**0.528**	**0.433**	**0.497**	**0.531**	0.542	0.565
	Random 50%	0.265	0.373	0.429	0.485	0.525	0.409	0.491	0.501	0.544	0.554
COCO	Unpruned	0.374	0.371	0.376	0.379	0.377	0.422	0.415	0.438	0.442	0.439
	Field H1 10%	0.288	0.302	0.310	0.330	0.339	0.370	0.371	**0.399**	**0.413**	0.415
	Field H0 10%	**0.297**	**0.310**	0.316	0.333	0.340	0.368	**0.374**	0.398	0.412	0.416
	Field H0+H1 10%	0.295	0.309	**0.318**	**0.334**	**0.341**	**0.371**	0.370	**0.399**	0.412	**0.417**
	L2 10%	0.273	0.293	0.303	0.323	0.334	0.349	0.356	0.384	0.403	0.409
	Random 10%	0.262	0.281	0.295	0.318	0.329	0.336	0.347	0.379	0.400	0.407
	Field H1 30%	0.128	**0.171**	0.189	**0.234**	**0.258**	0.212	**0.254**	0.294	0.336	0.352
	Field H0 30%	0.120	0.162	0.183	0.229	0.253	**0.215**	0.253	**0.299**	**0.337**	0.352
	Field H0+H1 30%	**0.129**	**0.171**	**0.192**	**0.234**	**0.258**	0.212	0.250	0.296	0.333	**0.354**
	L2 30%	0.072	0.112	0.142	0.193	0.227	0.155	0.206	0.252	0.301	0.326
	Random 30%	0.044	0.094	0.120	0.185	0.222	0.127	0.184	0.237	0.294	0.322
	Field H1 50%	**0.030**	**0.060**	**0.077**	**0.133**	**0.173**	**0.079**	**0.122**	**0.163**	**0.230**	0.266
	Field H0 50%	0.023	0.054	0.072	0.125	0.163	0.071	0.119	0.157	0.226	0.266
	Field H0+H1 50%	0.027	0.057	0.076	0.130	0.166	0.077	**0.122**	0.162	0.229	**0.267**
	L2 50%	0.010	0.028	0.039	0.081	0.120	0.037	0.068	0.108	0.172	0.222
	Random 50%	0.002	0.010	0.022	0.063	0.095	0.021	0.062	0.094	0.172	0.221

mAP50-95 after separate fine-tunes of 250 to 5,000 batches (WSD, Viewsphere: mean of 3 seeds; COCO: 1 seed on the 10k subset). One epoch is 63 / 58 / 625 batches. **Bold** = best criterion at that ratio and budget.

Table 6: COCO: recovery with longer fine-tunes

		YOLO26n				YOLO26s			
Data	Run	25,000	50,000	full 3 ep	full 7 ep	25,000	50,000	full 3 ep	full 7 ep
COCO	Unpruned	0.386	0.392	0.384	0.388	0.442	0.459	0.450	0.460
	Field H1 10%	0.339	**0.336**	0.361	0.371	0.396	**0.403**	**0.439**	0.448
	Field H0 10%	**0.340**	**0.336**	0.363	0.371	**0.399**	0.402	0.435	**0.449**
	Field H0+H1 10%	0.339	**0.336**	**0.364**	**0.372**	0.397	**0.403**	0.436	**0.449**
	L2 10%	0.336	0.333	0.359	0.371	0.392	0.384	0.432	**0.449**
	Random 10%	0.338	0.333	0.357	0.369	0.392	0.384	0.433	0.446
	Field H1 30%	0.288	0.290	0.311	0.330	**0.361**	0.353	0.399	0.421
	Field H0 30%	0.289	**0.293**	0.310	**0.331**	0.360	**0.354**	**0.401**	**0.422**
	Field H0+H1 30%	**0.292**	0.292	**0.314**	**0.331**	**0.361**	**0.354**	**0.401**	**0.422**
	L2 30%	0.277	0.282	0.299	0.323	0.350	0.346	0.388	0.413
	Random 30%	0.277	0.285	0.298	0.322	0.350	0.345	0.390	0.416
	Field H1 50%	**0.237**	**0.247**	**0.256**	**0.283**	0.307	0.308	**0.350**	0.380
	Field H0 50%	0.230	0.243	0.250	0.277	0.309	0.308	0.346	**0.381**
	Field H0+H1 50%	0.229	0.242	0.251	0.280	**0.310**	**0.311**	0.347	0.380
	L2 50%	0.205	0.226	0.220	0.260	0.288	0.296	0.327	0.363
	Random 50%	0.183	0.211	0.200	0.247	0.293	0.296	0.327	0.369

mAP50-95 on val2017 (1 seed). 25,000 and 50,000 batches on the 10k subset; *full 3 / 7 ep* = fine-tuned on all of train2017 (7,393 batches per epoch, so 22k / 52k batches). Bold as in Table 5.

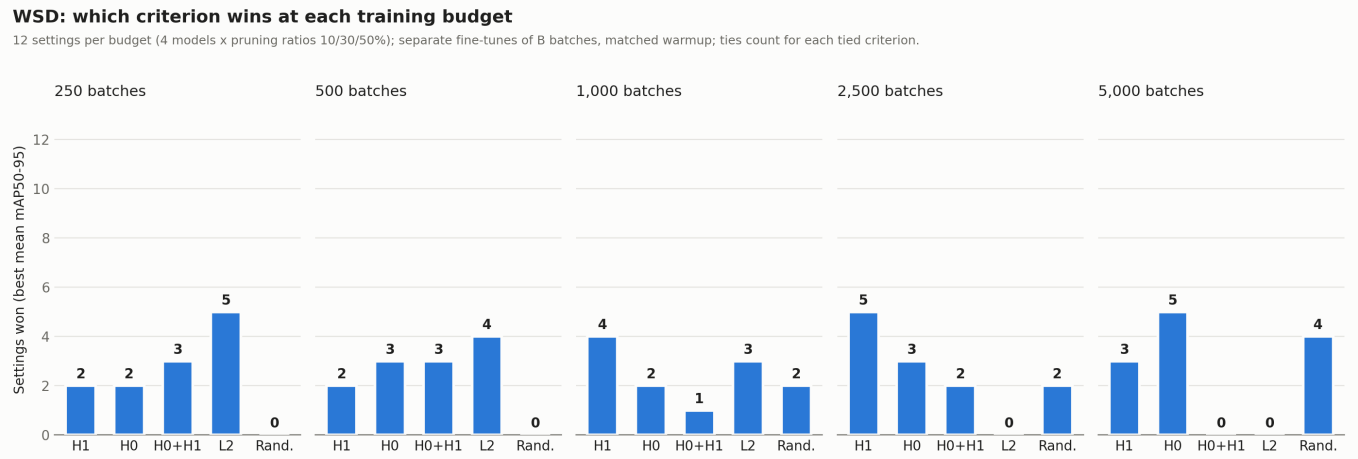
Table 7: Training needed to reach 95% of the unpruned model

		YOLO26n			YOLO26s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	500	5,000	**5,000**	**250**	**250**	**500**
	Field H0	**250**	2,500	**5,000**	**250**	2,500	never
	Field H0+H1	**250**	2,500	never	**250**	**250**	5,000
	L2	**250**	**1,000**	never	**250**	500	never
	Random	500	never	never	**250**	500	2,500
Viewsphere	Field H1	500	**5,000**	never	**250**	1,000	5,000
	Field H0	**250**	never	never	**250**	**500**	5,000
	Field H0+H1	**250**	never	never	500	1,000	**2,500**
	L2	**250**	**5,000**	never	**250**	1,000	5,000
	Random	500	**5,000**	never	500	2,500	5,000
COCO	Field H1	**7 ep (full)**	never	never	3 ep (full)	never	never
	Field H0	**7 ep (full)**	never	never	3 ep (full)	never	never
	Field H0+H1	**7 ep (full)**	never	never	**5,000**	never	never
	L2	**7 ep (full)**	never	never	3 ep (full)	never	never
	Random	**7 ep (full)**	never	never	3 ep (full)	never	never

Smallest training budget tested after which the pruned model reaches 95% of the unpruned model's mAP50-95 at the same budget (WSD, Viewsphere: 3-seed means). *never* = not within 5,000 batches (COCO: not within 7 epochs on all of train2017). **Bold** = fastest criterion. The budgets are coarse steps (250, 500, 1,000, 2,500, 5,000), so ties are common and a step can move with seed noise.

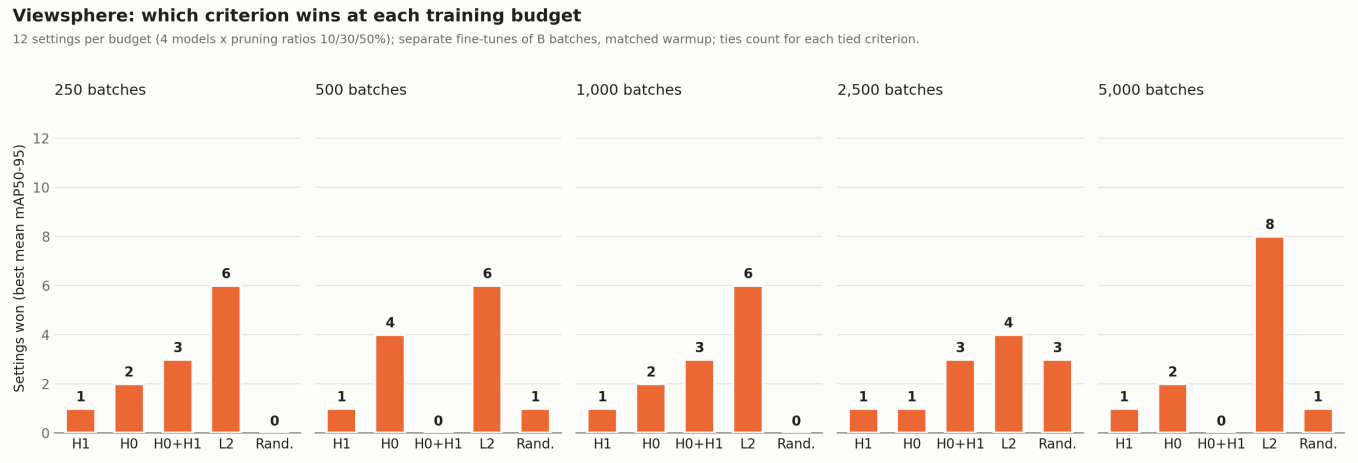
Figures 7–11 count, at each training budget, which criterion recovers best: one figure per dataset, then all four together. Both architectures are included (YOLOv5: Section 4; Viewsphere v2: Section 5). A setting = model (YOLO26n/s, YOLOv5n/s) × pruning ratio (10, 30, 50%), 12 per dataset and budget; the criterion with the highest mean mAP50-95 wins it (ties count for each), which is not necessarily a lead beyond the seed noise. COCO's longer budgets are left out.

Figure 7: WSD: which criterion wins at each training budget



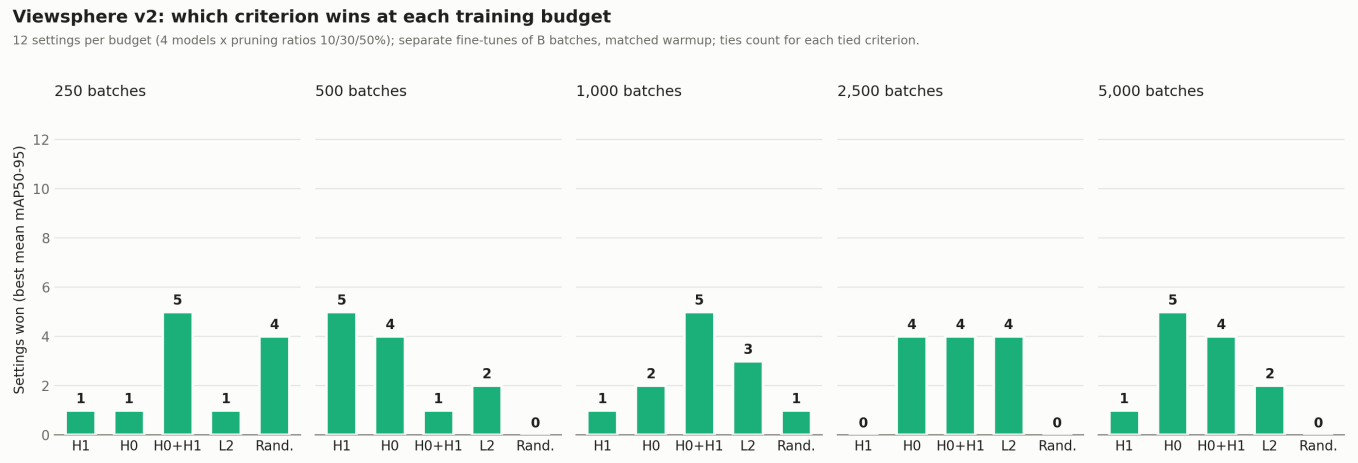
The winner changes with the budget: L2 at 250–500 batches, field H1 at 1,000–2,500, field H0 at 5,000, where random also wins 4 of 12; on WSD the criteria end close together.

Figure 8: Viewsphere: which criterion wins at each training budget



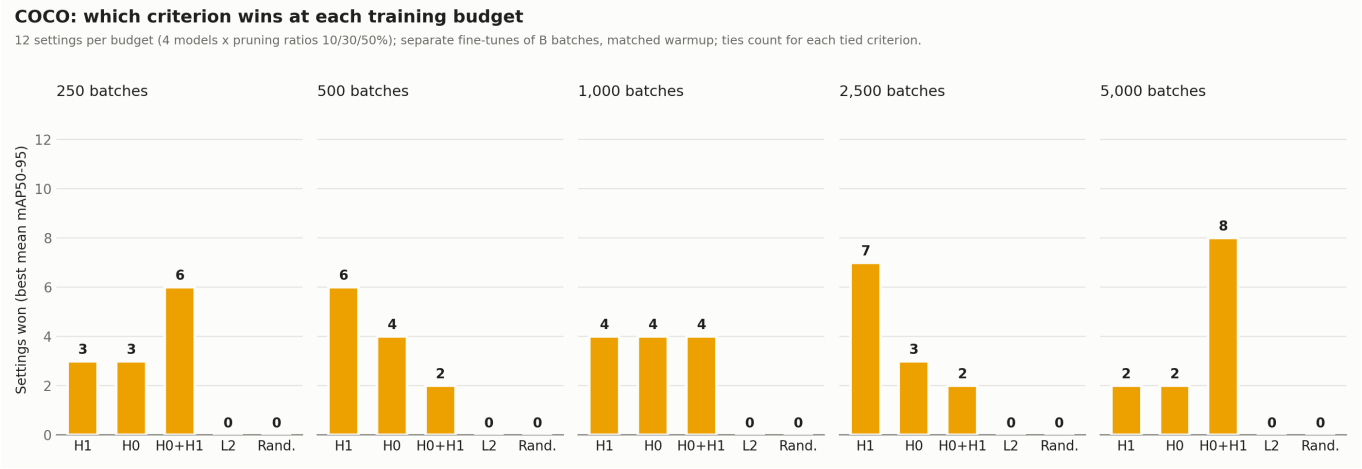
L2 wins most often at every budget (4–8 of 12).

Figure 9: Viewsphere v2: which criterion wins at each training budget



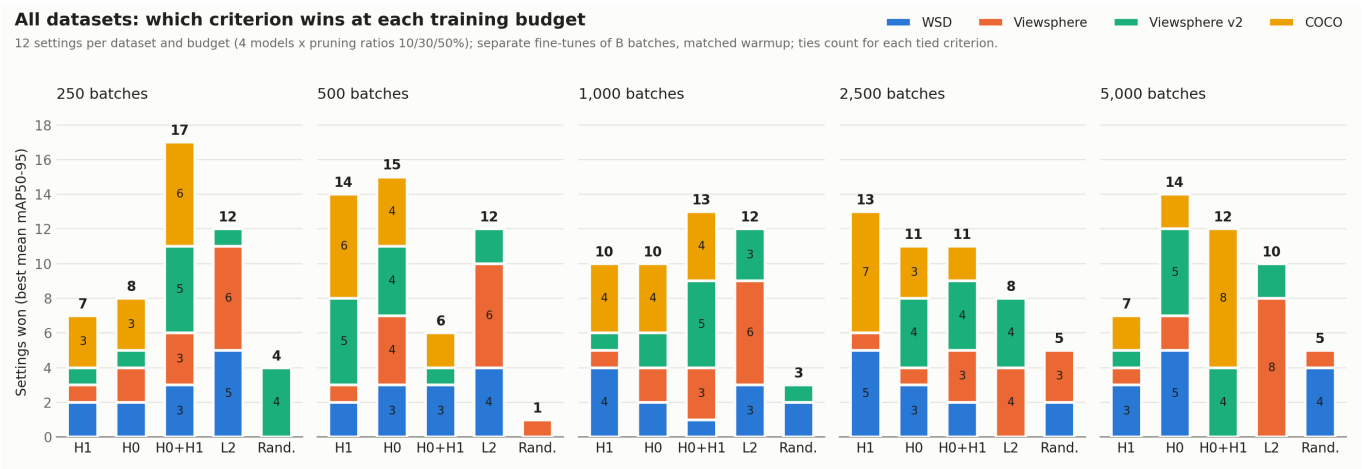
No consistent winner: field H0+H1 at 250 and 1,000 batches, field H1 at 500, field H0 at 5,000; the short budgets are affected by the BN problem (Section 5).

Figure 10: COCO: which criterion wins at each training budget



A persistence score wins every COCO setting at every budget; L2 and random never do. Field H1 leads at 500 and 2,500 batches, field H0+H1 at 250 and 5,000.

Figure 11: All datasets: which criterion wins at each training budget



Figures 7–11 stacked. No criterion leads at every budget: field H0+H1 wins most often at 250 batches (17 of 48), field H0 and field H1 at 500, field H1 at 2,500 and field H0 at 5,000. 42 of L2's 54 wins are on Viewsphere; random wins 1–5 of 48.

3.5 Block replacement, all datasets

Does the block criterion matter? Table 8 compares each TDA ranking with cutting the same number of blocks at random. Random choices already spread widely (sd 0.01–0.04 mAP50-95), so *which* blocks are cut matters. On every dataset at least one TDA ranking beats random choice by a margin that holds up across seeds: grid on WSD (+0.012 to +0.036), head on Viewsphere at $k = 2, 3$ (+0.013 to +0.018), null on COCO (up to +0.118 at $k = 3$ on YOLO26n). But the winning ranking differs by dataset, and the others can be clearly worse than random (null on Viewsphere YOLO26s: -0.072 at $k = 3$). The criterion matters; no single ranking is safe to pick in advance. Scoring blocks on H0 or H1 alone, instead of the blend, often changes the cut set and with it the result: H1 alone gives the most reliable ranking on WSD (null beats random in all six model/k cases), H0 alone on COCO YOLO26s; on Viewsphere neither rescues grid or null.

Table 8: Block pruning: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLO26n	Random (mean ± sd)	0.346 ± 0.009			0.329 ± 0.015			0.319 ± 0.023		
	Head (raw)	**−0.013**	**−0.013**	**−0.013**	−0.006	−0.006	−0.006	**−0.027**	**−0.027**	**−0.027**
	All, grid	**+0.012**	**+0.012**	**+0.006**	**+0.036**	**+0.036**	+0.001	**+0.023**	**+0.023**	**+0.023**
	All, null	**−0.011**	−0.004	**+0.012**	+0.008	−0.007	**+0.037**	**+0.015**	**−0.014**	**+0.036**
WSD YOLO26s	Random (mean ± sd)	0.371 ± 0.009			0.355 ± 0.010			0.351 ± 0.022		
	Head (raw)	**−0.010**	**−0.010**	**−0.010**	+0.003	+0.003	+0.003	**−0.024**	**−0.024**	**−0.024**
	All, grid	−0.003	−0.003	+0.002	**+0.027**	−0.002	+0.004	**+0.016**	**+0.016**	**+0.028**
	All, null	**+0.011**	−0.001	**+0.011**	+0.008	**−0.010**	**+0.025**	−0.008	−0.008	**+0.043**
Viewsphere YOLO26n	Random (mean ± sd)	0.539 ± 0.011			0.497 ± 0.018			0.456 ± 0.028		
	Head (raw)	+0.001	+0.001	+0.001	**+0.013**	**+0.013**	**+0.013**	**+0.018**	**+0.018**	**+0.018**
	All, grid	**−0.007**	**−0.007**	+0.006	−0.012	**−0.011**	−0.012	+0.000	−0.016	**−0.040**
	All, null	**−0.030**	**−0.030**	+0.005	−0.004	−0.004	−0.023	+0.002	+0.002	−0.014
Viewsphere YOLO26s	Random (mean ± sd)	0.564 ± 0.013			0.531 ± 0.023			0.497 ± 0.024		
	Head (raw)	+0.003	+0.003	+0.003	**+0.016**	**+0.016**	**+0.016**	+0.018	+0.018	+0.018
	All, grid	**−0.009**	**−0.009**	**−0.009**	−0.011	−0.011	−0.011	**−0.031**	**−0.031**	**−0.031**
	All, null	**−0.033**	**−0.033**	−0.003	**−0.023**	**−0.023**	−0.049	**−0.072**	−0.027	−0.031
COCO YOLO26n	Random (mean ± sd)	0.301 ± 0.034			0.229 ± 0.040			0.147 ± 0.033		
	Head (raw)	+0.008	+0.008	+0.008	+0.006	+0.013	+0.006	**+0.020**	**+0.020**	**+0.020**
	All, grid	+0.009	+0.008	+0.014	**+0.022**	**+0.022**	+0.002	**−0.030**	**−0.030**	**−0.036**
	All, null	**+0.043**	**+0.043**	**−0.057**	**+0.066**	**+0.047**	**−0.044**	**+0.118**	**+0.049**	+0.013
COCO YOLO26s	Random (mean ± sd)	0.363 ± 0.024			0.304 ± 0.033			0.229 ± 0.039		
	Head (raw)	**+0.011**	**+0.011**	**+0.011**	+0.012	**+0.020**	+0.012	**+0.035**	**+0.035**	**+0.035**
	All, grid	**+0.033**	+0.002	**+0.011**	**+0.032**	−0.003	**−0.021**	**−0.022**	**+0.040**	**+0.018**
	All, null	**+0.033**	**+0.033**	**−0.032**	+0.001	**+0.032**	+0.001	**+0.023**	**+0.063**	**+0.023**

First row of each group: mAP50-95 after cutting k random blocks (23 orders, seed 42) and fine-tuning 20 epochs. Other rows: each TDA ranking's cut (mean of 3 seeds) minus that random mean. **Bold** = the gap exceeds two standard errors, i.e. the TDA choice is reliably better (+) or worse (−) than a random one. Scores: C = blend of H0 and H1, H0 = H0 only, H1 = H1 only; equal entries mean the scores pick the same blocks.

Table 9: Block pruning: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLO26n (0.363)	Random: median (range)	0.347 (0.331–0.356)			0.329 (0.290–0.363)			0.324 (0.275–0.357)		
	Head (raw)	0.333	0.333	0.333	0.323	0.323	0.323	0.293	0.293	0.293
	All, grid	**0.358**†	**0.358**†	0.352	0.365†	0.365†	0.330	0.342	0.342	0.342
	All, null	0.336	0.342	**0.358**†	0.336	0.322	**0.365**†	0.335	0.306	**0.355**
WSD YOLO26s (0.382)	Random: median (range)	0.369 (0.361–0.388)			0.351 (0.334–0.379)			0.354 (0.311–0.393)		
	Head (raw)	0.361	0.361	0.361	0.358	0.358	0.358	0.326	0.326	0.326
	All, grid	0.368	0.368	0.373	**0.382**†	0.353	0.359	0.366	0.366	0.378
	All, null	**0.382**	0.370	**0.382**	0.363	0.345	0.380†	0.343	0.343	**0.394**†
Viewsphere YOLO26n (0.566)	Random: median (range)	0.542 (0.513–0.550)			0.498 (0.463–0.530)			0.458 (0.375–0.499)		
	Head (raw)	0.540	0.540	0.540	**0.510**	**0.510**	**0.510**	**0.474**	**0.474**	**0.474**
	All, grid	0.531	0.531	**0.544**	0.485	0.486	0.485	0.457	0.440	0.416
	All, null	0.509	0.509	0.543	0.493	0.493	0.474	0.458	0.458	0.442
Viewsphere YOLO26s (0.569)	Random: median (range)	0.564 (0.538–0.581)			0.526 (0.488–0.570)			0.494 (0.448–0.540)		
	Head (raw)	**0.568**	**0.568**	**0.568**	**0.547**	**0.547**	**0.547**	**0.514**	**0.514**	**0.514**
	All, grid	0.555	0.555	0.555	0.520	0.520	0.520	0.466	0.466	0.465
	All, null	0.531	0.531	0.561	0.508	0.508	0.482	0.424	0.470	0.466
COCO YOLO26n (0.369)	Random: median (range)	0.303 (0.245–0.346)			0.237 (0.165–0.296)			0.144 (0.090–0.201)		
	Head (raw)	0.309	0.309	0.309	0.235	0.242	0.235	0.167	0.167	0.167
	All, grid	0.310	0.309	0.315	0.251	0.251	0.231	0.117	0.117	0.111
	All, null	**0.344**	**0.344**	0.244	**0.295**	0.276	0.185	**0.265**†	0.195	0.160
COCO YOLO26s (0.423)	Random: median (range)	0.362 (0.329–0.399)			0.308 (0.243–0.353)			0.225 (0.159–0.289)		
	Head (raw)	0.374	0.374	0.374	0.317	0.324	0.317	0.264	0.264	0.264
	All, grid	**0.397**	0.365	0.374	**0.336**	0.301	0.283	0.208	0.269	0.248
	All, null	**0.397**	**0.397**	0.331	0.305	**0.336**	0.305	0.252	**0.292**†	0.252

mAP50-95 after cutting k blocks and fine-tuning 20 epochs. Each ranking's cut is the mean of 3 seeds; the baseline is the spread of 23 random block orders (seed 42); uncut mAP in brackets. Scores as in Table 8. **Bold** = best ranking and score at that k; † = above every random order. Rankings that pick the same blocks share one trained model.

3.6 Inference cost

Table 10: Size and inference time of channel-pruned models

		YOLO26n				YOLO26s			
Data	Pruned	GFLOPs	GPU ms	GPU img/s	CPU ms	GFLOPs	GPU ms	GPU img/s	CPU ms
WSD	none	5.70	9.12	2,462	44.4	22.36	9.17	1,356	101.6
	10%	5.28	9.02	2,188	46.4	20.76	9.41	1,232	103.7
	20%	4.94	9.23	2,230	44.8	19.39	9.52	1,242	101.1
	30%	4.63	**8.76**	2,305	43.4	18.11	9.38	1,290	94.0
	40%	4.35	8.79	2,415	43.0	17.00	9.26	1,293	91.7
	50%	**4.12**	8.77	**2,718**	**38.9**	**16.04**	**9.09**	**1,557**	**83.6**
Viewsphere	none	5.70	9.13	2,447	45.2	22.36	9.16	1,329	107.4
	10%	5.28	9.04	2,170	47.3	20.76	9.42	1,220	103.5
	20%	4.94	9.22	2,227	46.4	19.39	9.50	1,230	100.8
	30%	4.63	**8.77**	2,276	49.6	18.11	9.35	1,294	**95.7**
	40%	4.35	8.78	2,403	48.3	17.00	9.23	1,305	96.4
	50%	**4.12**	**8.77**	**2,658**	**42.1**	**16.04**	**9.11**	**1,529**	95.9
COCO	none	6.04	9.22	2,290	45.9	22.69	9.25	1,312	103.6
	10%	5.62	9.11	2,073	47.9	21.09	9.50	1,201	105.8
	20%	5.28	9.29	2,129	46.6	19.72	9.58	1,206	101.7
	30%	4.97	8.86	2,196	45.3	18.44	9.43	1,273	94.9
	40%	4.69	8.86	2,310	44.6	17.33	9.32	1,291	93.6
	50%	**4.46**	**8.84**	**2,512**	**40.5**	**16.37**	**9.17**	**1,503**	**85.7**

The same for every criterion at a given ratio. GPU: NVIDIA A100, FP16, batch-1 latency (ms) and batch-32 throughput (images/s); CPU: Intel Xeon, FP32, 8 threads, batch-1 latency (ms). All three datasets were timed in one run. **Bold** = best value for that dataset and model (lowest GFLOPs and latency, highest throughput).

Pruning 50% of channels removes 26–28% of GFLOPs but changes GPU batch-1 latency by only 1–4%: at this model size the GPU is limited by kernel launches, not arithmetic. Batch-32 throughput even drops at 10–20% before rising 9–15% at 50%. CPU latency improves mainly at 50% (12–18% on WSD and COCO; the Viewsphere CPU timings are noisier). Replacing a block, by contrast, saves 7–14% GPU latency for one block and 18–28% for three.

4 YOLOv5

The whole study was repeated on YOLOv5n/s, in Ultralytics' anchor-free version (YOLOv5u): the same datasets, criteria, seeds, fine-tunes and recovery budgets as for YOLO26. Starting models: COCO = Ultralytics' pretrained weights; WSD = trained from scratch like the YOLO26 models (best mAP50-95 0.370 / 0.405 for n / s); Viewsphere = those fine-tuned on it (0.539 / 0.614). YOLOv5 has no attention layers, so the same channel pruning removes more: at 50%, 57–61% of parameters and 35–41% of GFLOPs (YOLO26: 39% and 26–28%). Its replaceable blocks are the eight C3 blocks (2, 4, 6, 8 in the backbone; 13, 17, 20, 23 in the head). Tables 11–19 match Tables 2–10.

4.1 Channel pruning

Table 11: YOLOv5, WSD: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.362			0.365	
	Field H1	0.303	0.122	0.009	0.361	0.359	0.359
	Field H0	0.290	0.104	0.007	0.361	**0.360**	0.354
	Field H0+H1	0.311	0.102	0.009	**0.365**	0.359	0.357
	L2	**0.319**†	**0.164**†	**0.031**†	0.363	0.358	**0.360**
	Random	0.258	0.037	0.002	0.360	0.354	0.342
YOLOv5s	Unpruned		0.399			0.393	
	Field H1	0.334	0.123	0.011	0.388	0.378	0.377
	Field H0	**0.348**†	0.130	0.015	0.387	0.375	0.381
	Field H0+H1	0.345	0.123	0.012	**0.389**	**0.382**	**0.385**
	L2	0.343	**0.205**†	**0.052**†	0.385	0.373	0.368
	Random	0.306	0.066	0.003	0.383	0.380	0.385

As Table 2.

Table 12: YOLOv5, Viewsphere: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.501			0.492	
	Field H1	0.366	0.118	0.004	0.488	0.426	0.353
	Field H0	**0.391**	**0.148**†	**0.018**	0.493	0.386	0.360
	Field H0+H1	0.369	0.124	0.017	0.490	0.424	0.351
	L2	0.384	0.068	0.006	**0.494**	**0.448**	**0.362**
	Random	0.362	0.024	0.001	0.481	0.436	0.319
YOLOv5s	Unpruned		0.608			0.593	
	Field H1	**0.501**	**0.213**	**0.025**†	0.576	0.552	**0.478**
	Field H0	0.472	0.170	0.006	**0.583**	0.538	0.476
	Field H0+H1	0.487	0.194	0.011	0.573	**0.554**	0.471
	L2	0.415	0.119	0.004	0.576	0.544	0.476
	Random	0.483	0.113	0.002	0.575	0.536	0.446

As Table 2.

Table 13: YOLOv5, COCO: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.333			0.308	
	Field H1	0.141	**0.003**	0.000	0.281	0.195	0.108
	Field H0	0.144	0.002	0.000	**0.281**	**0.196**	**0.108**
	Field H0+H1	**0.148**†	0.003	0.000	0.281	0.196	0.108
	L2	0.105	0.003	0.000	0.274	0.182	0.095
	Random	0.110	0.001	**0.000**	0.277	0.185	0.099
YOLOv5s	Unpruned		0.416			0.374	
	Field H1	0.256	0.021	**0.001**†	0.355	0.301	0.210
	Field H0	0.264	0.024	0.000	0.355	0.303	0.208
	Field H0+H1	**0.267**†	**0.028**†	0.000	**0.356**	0.303	**0.212**
	L2	0.209	0.009	0.000	0.353	0.294	0.188
	Random	0.239	0.009	0.000	0.355	**0.304**	0.196

As Table 2.

- **WSD:** as on YOLO26, L2 is the best criterion one-shot (by 0.01–0.04 on n; on s from 30%), and after fine-tuning everything ends within about 0.01 of the unpruned model; only random at 50% on n falls behind (0.342 vs 0.354–0.360). On s even random catches up.
- **Viewsphere:** one-shot, the best persistence score beats L2 from 30% on (on s at every ratio: field H1 0.501 vs L2 0.415 at 10%), unlike YOLO26. After fine-tuning L2 leads on n at 30% (0.448 vs 0.386–0.426); at 50% every scored criterion is ahead of random (0.35–0.36 vs 0.319 on n, 0.47–0.48 vs 0.446 on s).
- **COCO:** the persistence scores are best again, but by less than on YOLO26 and with L2 no better than random: one-shot at 10%, 0.141–0.148 vs 0.105 (L2) and 0.110 (random) on n; after fine-tuning at 50%, 0.108 vs 0.095 and 0.099 (n) and 0.208–0.212 vs 0.188 and 0.196 (s). The three persistence scores give almost the same models after fine-tuning.

4.2 Recovery

Table 14: YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
WSD	Unpruned	0.357	0.364	0.368	0.367	0.370	0.376	0.387	0.391	0.401	0.400
	Field H1 10%	0.351	0.360	0.361	0.359	0.364	**0.378**	0.384	**0.390**	**0.395**	0.396
	Field H0 10%	**0.358**	0.362	**0.366**	0.363	**0.367**	0.376	0.383	**0.390**	0.389	0.395
	Field H0+H1 10%	**0.358**	**0.364**	0.365	**0.366**	0.366	0.370	0.380	0.382	0.392	0.396
	L2 10%	0.355	0.359	0.362	0.361	0.362	0.375	**0.385**	0.387	0.391	0.394
	Random 10%	0.354	0.361	0.360	0.363	0.362	0.362	0.380	0.384	0.393	**0.398**
	Field H1 30%	**0.341**	0.351	0.356	**0.361**	0.365	0.350	0.369	0.377	0.385	0.388
	Field H0 30%	**0.341**	**0.358**	**0.361**	0.360	**0.367**	0.347	0.360	0.366	0.377	**0.389**
	Field H0+H1 30%	0.338	0.351	0.357	**0.361**	0.365	0.351	0.369	0.374	0.385	0.388
	L2 30%	0.340	0.353	**0.361**	0.357	0.358	**0.366**	**0.373**	0.375	0.377	0.384
	Random 30%	0.328	0.345	0.353	**0.361**	0.361	0.357	0.371	**0.381**	**0.386**	0.388
	Field H1 50%	0.302	**0.342**	**0.358**	**0.367**	**0.371**	0.319	0.357	0.376	**0.385**	0.384
	Field H0 50%	0.286	0.331	0.355	0.365	0.369	0.336	**0.365**	0.377	**0.385**	0.379
	Field H0+H1 50%	0.296	0.329	0.351	0.363	0.368	0.333	0.364	0.376	0.383	0.385
	L2 50%	**0.312**	0.341	0.351	0.362	0.364	**0.344**	0.363	0.368	0.374	0.378
	Random 50%	0.255	0.308	0.333	0.353	0.358	0.316	0.357	**0.378**	0.381	**0.387**
Viewsphere	Unpruned	0.454	0.482	0.453	0.515	0.516	0.563	0.565	0.587	0.599	0.605
	Field H1 10%	0.426	0.482	0.454	0.513	**0.518**	**0.557**	0.558	0.565	0.588	0.601
	Field H0 10%	0.421	0.473	**0.490**	0.453	0.485	0.535	0.538	**0.579**	**0.592**	0.598
	Field H0+H1 10%	0.422	0.468	0.476	**0.516**	0.509	0.536	0.550	0.562	0.586	0.602
	L2 10%	**0.439**	**0.484**	0.478	0.493	0.512	0.551	0.551	**0.579**	0.591	**0.604**
	Random 10%	0.417	0.456	0.467	0.507	0.478	0.527	**0.559**	0.569	0.587	0.594
	Field H1 30%	0.355	0.376	0.410	0.463	0.491	0.494	**0.522**	0.546	0.527	0.583
	Field H0 30%	0.374	0.400	0.411	0.456	0.477	0.504	0.501	0.544	0.564	0.576
	Field H0+H1 30%	**0.402**	0.362	**0.441**	0.453	0.464	0.487	0.507	**0.547**	**0.572**	0.582
	L2 30%	0.390	**0.429**	**0.441**	**0.475**	**0.497**	**0.508**	0.505	0.541	0.568	0.584
	Random 30%	0.331	0.386	0.387	0.429	0.479	0.453	0.518	0.538	0.555	**0.585**
	Field H1 50%	0.227	0.293	0.336	**0.422**	0.425	0.295	0.415	0.464	0.522	0.551
	Field H0 50%	0.259	**0.323**	0.348	0.392	0.414	0.353	**0.419**	0.453	0.480	**0.552**
	Field H0+H1 50%	0.207	0.298	0.366	0.384	0.416	**0.355**	0.412	**0.479**	0.519	0.540
	L2 50%	**0.276**	0.310	**0.381**	0.415	**0.428**	0.211	0.418	0.421	**0.537**	**0.552**
	Random 50%	0.179	0.264	0.325	0.404	0.415	0.305	0.367	0.420	0.494	0.531
COCO	Unpruned	0.298	0.295	0.305	0.309	0.311	0.353	0.347	0.374	0.382	0.383
	Field H1 10%	0.211	**0.227**	0.250	**0.267**	0.277	**0.293**	0.292	**0.331**	**0.351**	0.357
	Field H0 10%	0.216	0.226	**0.251**	0.265	**0.278**	0.292	**0.298**	**0.331**	0.349	0.356
	Field H0+H1 10%	**0.218**	0.226	0.249	0.266	0.276	**0.293**	0.295	0.330	0.348	**0.360**
	L2 10%	0.196	0.209	0.235	0.255	0.266	0.272	0.283	0.322	0.343	0.354
	Random 10%	0.198	0.215	0.240	0.260	0.273	0.284	0.291	0.326	0.350	0.359
	Field H1 30%	0.046	**0.076**	0.109	**0.152**	**0.181**	0.125	0.157	0.206	0.256	0.285
	Field H0 30%	**0.047**	0.071	**0.110**	0.149	0.178	0.126	0.161	0.215	**0.263**	**0.290**
	Field H0+H1 30%	**0.047**	**0.076**	0.109	0.150	**0.181**	**0.131**	**0.166**	**0.216**	**0.263**	0.289
	L2 30%	0.034	0.061	0.088	0.132	0.164	0.089	0.126	0.183	0.244	0.278
	Random 30%	0.027	0.056	0.090	0.135	0.167	0.112	0.154	0.206	0.261	0.288
	Field H1 50%	0.006	0.016	0.029	0.058	0.085	0.023	0.047	0.086	0.141	0.180
	Field H0 50%	**0.008**	**0.019**	**0.033**	**0.062**	0.086	**0.029**	0.049	0.087	**0.145**	0.184
	Field H0+H1 50%	**0.008**	**0.019**	0.032	**0.062**	**0.089**	0.025	**0.053**	**0.088**	**0.145**	**0.187**
	L2 50%	0.004	0.012	0.027	0.051	0.074	0.014	0.031	0.066	0.117	0.165
	Random 50%	0.004	0.013	0.027	0.054	0.078	0.013	0.036	0.074	0.135	0.176

As Table 5.

Table 15: YOLOv5, COCO: recovery with longer fine-tunes

		YOLOv5n				YOLOv5s			
Data	Run	25,000	50,000	full 3 ep	full 7 ep	25,000	50,000	full 3 ep	full 7 ep
COCO	Unpruned	0.316	0.328	0.319	0.328	0.383	0.403	0.394	0.412
	Field H1 10%	**0.281**	**0.279**	**0.300**	**0.313**	**0.347**	**0.342**	0.381	0.401
	Field H0 10%	**0.281**	0.276	0.298	0.312	0.344	0.341	0.380	0.401
	Field H0+H1 10%	**0.281**	0.276	0.298	0.312	0.345	**0.342**	**0.382**	**0.402**
	L2 10%	0.275	0.273	0.293	0.308	0.345	0.338	0.378	0.398
	Random 10%	0.280	0.275	0.296	0.311	0.343	0.337	0.379	0.400
	Field H1 30%	**0.227**	0.234	**0.243**	0.262	0.308	0.303	0.338	0.368
	Field H0 30%	0.224	0.234	0.241	0.263	**0.309**	0.307	**0.343**	**0.369**
	Field H0+H1 30%	0.224	**0.236**	0.239	**0.264**	0.308	0.304	0.341	0.368
	L2 30%	0.218	0.230	0.232	0.257	0.305	0.304	0.332	0.361
	Random 30%	0.223	0.234	0.237	0.261	**0.309**	**0.308**	0.340	0.367
	Field H1 50%	0.147	0.168	0.159	0.191	0.247	0.258	0.268	0.309
	Field H0 50%	0.147	**0.170**	0.158	0.190	**0.255**	**0.264**	**0.272**	0.309
	Field H0+H1 50%	**0.150**	0.169	**0.160**	**0.192**	0.250	0.262	0.270	**0.311**
	L2 50%	0.138	0.164	0.149	0.186	0.241	0.256	0.260	0.301
	Random 50%	0.143	0.165	0.150	0.186	0.249	0.263	0.268	0.309

As Table 6.

Table 16: YOLOv5: training needed to reach 95% of the unpruned model

		YOLOv5n			YOLOv5s		
Data	Criterion	10%	30%	50%	10%	30%	50%
WSD	Field H1	**250**	**250**	**1,000**	**250**	500	**1,000**
	Field H0	**250**	**250**	**1,000**	**250**	5,000	**1,000**
	Field H0+H1	**250**	500	**1,000**	**250**	500	**1,000**
	L2	**250**	**250**	**1,000**	**250**	**250**	never
	Random	**250**	1,000	2,500	**250**	500	**1,000**
Viewsphere	Field H1	500	5,000	never	**250**	5,000	never
	Field H0	500	never	never	**250**	5,000	never
	Field H0+H1	500	**1,000**	never	**250**	**2,500**	never
	L2	**250**	**1,000**	never	**250**	5,000	never
	Random	1,000	never	never	500	5,000	never
COCO	Field H1	**7 ep (full)**	never	never	**3 ep (full)**	never	never
	Field H0	never	never	never	**3 ep (full)**	never	never
	Field H0+H1	never	never	never	**3 ep (full)**	never	never
	L2	never	never	never	**3 ep (full)**	never	never
	Random	never	never	never	**3 ep (full)**	never	never

As Table 7.

On COCO the best persistence score recovers fastest at 50% at every budget, but the margins are small (0.001–0.02; after 7 epochs on all of train2017, 0.190–0.192 vs 0.186 on n). YOLOv5 recovers more slowly than YOLO26 on COCO: even 10% pruning reaches 95% of the unpruned model only on s after 3 full epochs (and on n only with field H1, after 7). On WSD, 50% pruning is back to 95% within 1,000 batches with every persistence score (random: 2,500 on n; L2: never on s); on Viewsphere 50% never gets there within 5,000 batches.

4.3 Block replacement

Table 17: YOLOv5 block pruning: TDA-chosen blocks vs randomly chosen blocks

Data, model	Ranking	$k=1$			$k=2$			$k=3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLOv5n	Random (mean ± sd)	0.355 ± 0.008			0.346 ± 0.011			0.338 ± 0.015		
	Head (raw)	+0.002	+0.002	+0.002	−0.000	−0.000	−0.000	+0.006	+0.006	+0.006
	All, grid	**+0.011**	**+0.011**	+0.002	**+0.014**	**+0.014**	+0.005	**+0.018**	**+0.015**	**+0.018**
	All, null	+0.001	**−0.008**	**+0.011**	**+0.017**	**−0.018**	**+0.017**	**−0.011**	**−0.026**	**+0.015**
WSD YOLOv5s	Random (mean ± sd)	0.379 ± 0.009			0.368 ± 0.012			0.364 ± 0.015		
	Head (raw)	**−0.007**	**−0.007**	**−0.007**	+0.009	+0.009	+0.009	+0.003	+0.003	+0.003
	All, grid	+0.006	+0.006	+0.006	**+0.018**	**+0.018**	**+0.018**	**+0.017**	**+0.017**	**+0.017**
	All, null	−0.002	−0.007	−0.002	**+0.018**	**−0.013**	**+0.018**	−0.001	**−0.032**	**+0.026**
Viewsphere YOLOv5n	Random (mean ± sd)	0.496 ± 0.020			0.461 ± 0.032			0.422 ± 0.043		
	Head (raw)	**+0.022**	**+0.022**	**+0.022**	−0.003	+0.001	−0.003	**+0.036**	**+0.036**	**+0.036**
	All, grid	**+0.010**	**+0.010**	**−0.042**	**−0.030**	+0.014	**−0.057**	**−0.061**	**−0.061**	**−0.038**
	All, null	**−0.042**	−0.010	−0.010	**−0.057**	**−0.057**	−0.026	**−0.050**	**−0.038**	**−0.062**
Viewsphere YOLOv5s	Random (mean ± sd)	0.566 ± 0.016			0.536 ± 0.029			0.491 ± 0.049		
	Head (raw)	+0.006	+0.006	+0.006	**+0.027**	**+0.027**	**+0.027**	**+0.038**	**+0.038**	**+0.038**
	All, grid	−0.004	−0.004	−0.004	+0.008	+0.008	**+0.019**	**−0.033**	**−0.033**	**−0.033**
	All, null	**−0.020**	−0.018	+0.008	**−0.054**	**−0.023**	**+0.032**	**−0.049**	**−0.049**	+0.005
COCO YOLOv5n	Random (mean ± sd)	0.253 ± 0.024			0.207 ± 0.037			0.158 ± 0.033		
	Head (raw)	**+0.025**	**+0.025**	**+0.025**	**+0.041**	**+0.041**	**+0.041**	**+0.051**	**+0.051**	**+0.051**
	All, grid	**+0.025**	**+0.025**	**+0.025**	**+0.028**	+0.005	**+0.028**	+0.007	+0.007	**+0.051**
	All, null	**+0.021**	**+0.034**	**+0.025**	**+0.028**	**+0.050**	+0.005	**+0.051**	**+0.083**	+0.007
COCO YOLOv5s	Random (mean ± sd)	0.331 ± 0.016			0.288 ± 0.034			0.238 ± 0.042		
	Head (raw)	**+0.023**	**+0.023**	**+0.023**	**+0.041**	**+0.041**	**+0.030**	**+0.053**	**+0.053**	**+0.053**
	All, grid	**+0.023**	+0.002	**+0.023**	**+0.030**	**+0.029**	**+0.030**	**+0.019**	**+0.044**	**+0.019**
	All, null	**+0.011**	**+0.011**	**+0.023**	**+0.030**	**+0.036**	**+0.022**	**+0.054**	**+0.031**	**+0.019**

As Table 8.

Table 18: YOLOv5 block pruning: accuracy of each ranking's cut

Data, model (uncut)	Ranking	$k=1$			$k=2$			$k=3$		
		C	H0	H1	C	H0	H1	C	H0	H1
WSD YOLOv5n (0.366)	Random: median (range)	0.356 (0.344–0.367)			0.344 (0.327–0.364)			0.342 (0.317–0.360)		
	Head (raw)	0.357	0.357	0.357	0.346	0.346	0.346	0.344	0.344	0.344
	All, grid	**0.366**	**0.366**	0.357	0.361	0.361	0.352	**0.355**	0.353	**0.355**
	All, null	0.356	0.347	**0.366**	**0.363**	0.329	**0.363**	0.327	0.312	0.353
WSD YOLOv5s (0.389)	Random: median (range)	0.381 (0.366–0.391)			0.367 (0.345–0.386)			0.365 (0.334–0.389)		
	Head (raw)	0.372	0.372	0.372	0.377	0.377	0.377	0.367	0.367	0.367
	All, grid	**0.385**	**0.385**	**0.385**	**0.385**	**0.385**	**0.385**	0.382	0.382	0.382
	All, null	0.377	0.372	0.377	**0.385**	0.355	**0.385**	0.363	0.332	**0.390**†
Viewsphere YOLOv5n (0.522)	Random: median (range)	0.506 (0.457–0.509)			0.464 (0.391–0.504)			0.424 (0.340–0.471)		
	Head (raw)	**0.518**†	**0.518**†	**0.518**†	0.458	0.462	0.458	**0.458**	**0.458**	**0.458**
	All, grid	0.506	0.506	0.454	0.431	**0.474**	0.404	0.360	0.360	0.384
	All, null	0.454	0.486	0.486	0.404	0.404	0.434	0.372	0.384	0.360
Viewsphere YOLOv5s (0.589)	Random: median (range)	0.565 (0.539–0.591)			0.541 (0.493–0.575)			0.504 (0.370–0.565)		
	Head (raw)	0.572	0.572	0.572	0.563	0.563	0.563	**0.529**	**0.529**	**0.529**
	All, grid	0.562	0.562	0.562	0.544	0.544	0.555	0.458	0.458	0.458
	All, null	0.546	0.548	**0.574**	0.481	0.513	**0.568**	0.442	0.442	0.495
COCO YOLOv5n (0.308)	Random: median (range)	0.245 (0.218–0.287)			0.211 (0.142–0.263)			0.158 (0.109–0.220)		
	Head (raw)	0.278	0.278	0.278	0.248	0.248	0.248	0.209	0.209	0.209
	All, grid	0.278	0.278	0.278	0.235	0.212	0.235	0.166	0.166	0.209
	All, null	0.274	**0.287**†	0.278	0.235	**0.257**	0.212	0.209	**0.241**†	0.166
COCO YOLOv5s (0.374)	Random: median (range)	0.329 (0.308–0.356)			0.304 (0.224–0.335)			0.253 (0.159–0.305)		
	Head (raw)	**0.354**	**0.354**	**0.354**	**0.329**	**0.329**	0.318	0.291	0.291	0.291
	All, grid	**0.354**	0.334	**0.354**	0.318	0.317	0.318	0.257	0.282	0.257
	All, null	0.342	0.342	**0.354**	0.318	0.323	0.310	**0.292**	0.269	0.257

As Table 9.

The pattern by dataset is the same as on YOLO26, which suggests it comes from the data rather than the network: the size-normalised grid ranking beats random choice on WSD (+0.011 to +0.018 where it holds up), the raw head ranking does on Viewsphere (+0.027 and +0.038 at $k = 2, 3$ on s; +0.022 and +0.036 at $k = 1, 3$ on n) while grid and null are mostly worse than random there, and on COCO most rankings beat random by a margin that holds up (up to +0.083: null scored on H0 alone, $k = 3$ on n). Scoring on H0 alone hurts the null ranking on WSD (−0.008 to −0.032), as it did on YOLO26.

4.4 Inference cost

Table 19: YOLOv5: size and inference time of channel-pruned models

		YOLOv5n				YOLOv5s			
Data	Pruned	GFLOPs	GPU ms	GPU img/s	CPU ms	GFLOPs	GPU ms	GPU img/s	CPU ms
WSD	none	7.13	5.93	2,477	36.1	23.94	**5.82**	1,498	78.9
	10%	6.46	6.03	2,330	38.3	21.58	6.07	1,397	80.8
	20%	5.91	6.07	2,340	37.0	19.44	6.08	1,477	76.5
	30%	5.36	6.08	2,505	34.5	17.50	6.08	1,538	70.1
	40%	4.90	6.11	2,513	33.9	15.74	6.06	1,633	66.0
	50%	**4.49**	**5.91**	**2,901**	**29.6**	**14.24**	5.89	**1,854**	**58.4**
Viewsphere	none	7.13	5.92	2,458	36.7	23.94	**5.82**	1,483	82.2
	10%	6.46	6.03	2,346	39.0	21.58	6.08	1,388	81.1
	20%	5.91	6.07	2,344	37.1	19.44	6.09	1,475	77.2
	30%	5.36	6.08	2,478	34.7	17.50	6.08	1,542	71.1
	40%	4.90	6.12	2,508	33.9	15.74	6.05	1,638	66.7
	50%	**4.49**	**5.91**	**2,873**	**29.6**	**14.24**	5.88	**1,849**	**59.1**
COCO	none	7.79	5.95	2,378	37.7	24.11	**5.85**	1,482	79.6
	10%	7.10	6.06	2,231	40.1	21.75	6.11	1,373	81.8
	20%	6.53	6.09	2,203	38.7	19.61	6.11	1,452	77.4
	30%	5.96	6.11	2,368	36.2	17.66	6.11	1,518	71.1
	40%	5.47	6.14	2,410	35.5	15.90	6.10	1,594	66.7
	50%	**5.04**	**5.95**	**2,685**	**31.1**	**14.40**	5.93	**1,824**	**59.2**

As Table 10; timed in the same run as the YOLO26 models.

YOLOv5 runs about a third faster than YOLO26 at batch 1 on the GPU (5.9 vs 9.1 ms unpruned), despite having more GFLOPs. As on YOLO26, channel pruning barely moves GPU batch-1 latency: removing 35–41% of GFLOPs at 50% leaves it at 5.9 ms, and 10–40% pruning is slightly slower than none. Batch-32 throughput rises 17–24% at 50% and CPU latency falls 18–26%, more than on YOLO26 because more of the network is pruned. Replacing blocks saves 5–11% GPU latency for one block and 15–23% for three.

5 Viewsphere v2

Viewsphere v2 renders 17 satellites (the 4 of v1 plus AcrimSAT, CALIPSO, CloudSat B and C, a 2U CubeSat, EO-1, GRACE, ICESat-2, Landsat 5 and 8, Jason-2, QuikSCAT and TDRS-D), relabelled to WSD's 4 classes and split by viewing angle exactly as v1 (4,079 / 680 / 680 train / val / test images). The whole study was run on it for both architectures, with v1's settings (lr 1.25×10^{-3}, matched warmup bias lr; budgets 250–5,000 batches, here 1–20 epochs) but without the activation-basis grid. Starting models = the WSD models fine-tuned on v2, best mAP50-95 0.445 / 0.515 (YOLO26n/s) and 0.409 / 0.489 (YOLOv5n/s), lower than on v1 (0.54–0.61).

BN re-estimation is unreliable on v2. Re-estimating the batch-norm statistics on the *unpruned* model, the step every one-shot number includes, collapses mAP50-95 for some calibration seeds: YOLO26n gives 0.09 for seeds 42 and 46 but 0.42–0.45 for 43–45; YOLOv5n 0.06–0.22 for three of five seeds; YOLOv5s 0.27–0.36 for three. On v1 the same step costs at most about 0.09. The one-shot columns below (seed 42 for the unpruned row) are therefore mostly calibration luck, and only the fine-tuned columns compare the criteria. Fine-tuning repairs it: every unpruned seed ends at 0.45–0.47 (YOLO26n).

5.1 Channel pruning

Table 20: Viewsphere v2, YOLO26: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLO26n	Unpruned		0.089			0.460	
	Field H1	0.226	0.074	**0.014**	0.444	0.422	0.381
	Field H0	**0.235**	0.078	0.013	0.448	**0.425**	0.385
	Field H0+H1	0.224	**0.089**	0.013	0.446	0.417	0.376
	L2	0.192	0.034	0.002	**0.451**	0.419	**0.390**
	Random	0.233	0.030	0.003	0.431	0.381	0.370
YOLO26s	Unpruned		0.509			0.492	
	Field H1	0.374	**0.200**	**0.062**	0.484	0.462	0.431
	Field H0	0.382	0.168	0.048	0.483	0.464	0.427
	Field H0+H1	0.370	0.189	0.055	0.482	0.458	0.434
	L2	0.352	0.101	0.003	**0.488**	**0.471**	**0.434**
	Random	**0.399**	0.160	0.012	0.487	0.460	0.427

As Table 2. One-shot numbers include the unreliable BN re-estimation (see above).

Table 21: Viewsphere v2, YOLOv5: channel pruning by criterion

		One-shot			After 20-epoch fine-tune		
Model	Criterion	10%	30%	50%	10%	30%	50%
YOLOv5n	Unpruned		0.074			0.404	
	Field H1	0.171	0.049	0.003	0.373	0.344	0.153
	Field H0	**0.204**	**0.067**	0.004	0.370	0.338	0.307
	Field H0+H1	0.196	0.067	**0.004**	**0.395**	0.336	0.227
	L2	0.191	0.015	0.002	0.348	**0.373**	**0.313**
	Random	0.155	0.011	0.001	0.346	0.362	0.244
YOLOv5s	Unpruned		0.294			0.453	
	Field H1	0.304	0.125	0.010	0.404	0.374	**0.385**
	Field H0	0.317	0.114	0.009	0.419	**0.433**	0.330
	Field H0+H1	0.306	**0.142**	**0.010**	**0.436**	0.389	0.366
	L2	**0.323**	0.053	0.002	0.418	0.348	0.331
	Random	0.222	0.038	0.001	0.429	0.402	0.317

As Table 20.

- **YOLO26:** after fine-tuning, all criteria end within 0.02 of each other at 50% (0.376–0.390 on n, 0.427–0.434 on s). L2 has the highest (or tied) mean at 50% on both models and at 30% on s, but no lead exceeds two standard errors (no † anywhere in the table). Random is the worst on n at every ratio (0.370–0.431).
- **YOLOv5:** at 50% the fine-tunes are unstable: single seeds collapse (field H1 on n: 0.03–0.31 across seeds; L2 on s: 0.17–0.40), so means differ by up to 0.16 without any of them holding up. On s, the persistence scores are ahead of random at 50% (0.330–0.385 vs 0.317); on n, field H0 and L2 are the only criteria that stay near 0.31.

5.2 Recovery

Table 22: Viewsphere v2, YOLO26: recovery after channel pruning, by training budget

		YOLO26n					YOLO26s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.057	0.441	0.429	0.451	0.462	0.300	0.473	0.480	0.483	0.489
	Field H1 10%	0.053	**0.427**	0.416	0.438	0.434	0.346	0.465	0.473	0.469	0.485
	Field H0 10%	0.023	0.425	0.423	0.428	0.441	**0.424**	**0.466**	0.475	0.478	0.485
	Field H0+H1 10%	**0.137**	0.412	0.421	**0.446**	**0.447**	0.304	0.461	0.468	0.475	**0.490**
	L2 10%	0.061	0.426	**0.424**	0.440	0.445	0.235	**0.466**	**0.482**	**0.484**	0.484
	Random 10%	0.035	0.413	0.414	0.429	0.445	0.355	0.456	0.465	0.474	0.488
	Field H1 30%	0.015	0.227	0.379	0.405	0.414	**0.319**	**0.421**	0.424	0.444	0.455
	Field H0 30%	0.029	**0.372**	**0.382**	**0.408**	**0.431**	0.267	0.414	**0.432**	0.452	0.462
	Field H0+H1 30%	**0.052**	0.349	0.365	0.407	0.415	0.316	0.405	0.426	**0.453**	**0.466**
	L2 30%	0.023	0.336	0.363	0.404	0.422	0.157	0.410	0.426	0.450	**0.466**
	Random 30%	0.007	0.323	0.350	0.390	0.412	0.160	0.406	**0.432**	0.451	0.436
	Field H1 50%	0.012	**0.265**	0.292	0.348	0.367	0.060	0.189	0.372	0.407	0.430
	Field H0 50%	0.040	0.214	0.293	0.358	0.322	0.066	0.330	0.373	**0.414**	0.435
	Field H0+H1 50%	0.026	0.250	**0.308**	0.348	**0.390**	**0.225**	**0.353**	**0.380**	0.411	**0.438**
	L2 50%	0.010	0.149	0.290	**0.362**	0.387	0.061	0.273	0.366	0.411	0.431
	Random 50%	**0.073**	0.118	0.249	0.334	0.364	0.045	0.300	0.345	0.389	0.432

As Table 5.

Table 23: Viewsphere v2, YOLOv5: recovery after channel pruning, by training budget

		YOLOv5n					YOLOv5s				
Data	Run	250	500	1,000	2,500	5,000	250	500	1,000	2,500	5,000
Viewsphere v2	Unpruned	0.156	0.272	0.260	0.397	0.415	0.233	0.236	0.399	0.416	0.430
	Field H1 10%	0.036	0.262	0.325	0.368	0.368	0.231	0.256	0.342	0.420	0.390
	Field H0 10%	0.144	**0.332**	0.350	**0.395**	**0.393**	0.259	**0.270**	0.368	0.406	0.391
	Field H0+H1 10%	0.106	0.170	**0.375**	0.382	0.372	**0.343**	0.193	**0.397**	**0.444**	0.428
	L2 10%	0.200	0.309	0.313	0.384	0.375	0.207	0.156	0.277	0.321	**0.452**
	Random 10%	**0.230**	0.201	0.327	0.362	0.356	0.096	0.246	0.302	0.356	0.440
	Field H1 30%	0.126	**0.331**	0.254	0.339	0.197	0.109	**0.329**	0.302	0.365	0.397
	Field H0 30%	0.121	0.275	0.243	0.319	**0.374**	0.169	0.301	0.219	0.230	**0.439**
	Field H0+H1 30%	**0.169**	0.239	0.281	0.289	0.357	0.120	0.231	**0.325**	0.303	0.384
	L2 30%	0.113	0.257	**0.303**	**0.350**	0.363	**0.317**	0.290	0.231	**0.380**	0.355
	Random 30%	0.101	0.225	0.229	0.342	0.339	0.126	0.304	0.255	0.359	0.424
	Field H1 50%	0.013	0.180	0.151	0.253	0.274	0.054	0.231	**0.272**	0.310	**0.367**
	Field H0 50%	0.092	0.137	**0.254**	0.268	**0.291**	0.146	**0.240**	0.231	**0.315**	0.339
	Field H0+H1 50%	0.039	0.159	0.159	**0.310**	0.264	0.097	0.155	0.235	0.237	0.340
	L2 50%	0.091	**0.188**	0.248	0.241	0.290	0.173	0.098	0.219	0.242	0.304
	Random 50%	**0.167**	0.096	0.214	0.246	0.254	**0.181**	0.107	0.181	0.186	0.335

As Table 5.

The short budgets inherit the BN problem: the unpruned YOLO26n scores 0.057 after 250 batches and 0.44 after 500, and YOLOv5 does not improve steadily with budget (YOLOv5n unpruned: 0.272 at 500, 0.260 at 1,000). A "batches to 95% of unpruned" table like Table 7 would mostly measure that, so it is left out. By 5,000 batches (20 epochs), YOLO26 at 50% is 0.07–0.14 behind the unpruned model on n and 0.05–0.06 on s, with field H0+H1 the highest on both (0.390 and 0.438); YOLOv5 at 50% is 0.12–0.16 behind on n and 0.06–0.13 on s.

5.3 Block replacement

Table 24: Viewsphere v2 block pruning: TDA-chosen blocks vs randomly chosen blocks, YOLO26

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLO26n	Random (mean ± sd)	0.443 ± 0.006			0.380 ± 0.098			0.364 ± 0.056		
	Head (raw)	+0.001	+0.001	+0.001	+0.031	+0.031	+0.031	**+0.033**	**+0.033**	**+0.033**
	All, grid	−0.002	−0.002	−0.002	−0.095	−0.095	+0.028	+0.018	−0.025	+0.018
	All, null	−0.003	−0.003	−0.009	+0.038	+0.026	+0.036	**+0.047**	+0.023	+0.003
Viewsphere v2 YOLO26s	Random (mean ± sd)	0.477 ± 0.009			0.455 ± 0.016			0.421 ± 0.025		
	Head (raw)	+0.007	+0.007	+0.007	+0.004	+0.004	+0.000	+0.008	+0.008	+0.008
	All, grid	−0.036	−0.036	−0.036	−0.022	−0.022	−0.000	−0.017	−0.017	−0.017
	All, null	**−0.018**	**−0.031**	**+0.008**	**−0.021**	**−0.021**	−0.002	−0.019	−0.064	**−0.028**

As Table 8.

Table 25: Viewsphere v2 block pruning: TDA-chosen blocks vs randomly chosen blocks, YOLOv5

Data, model	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLOv5n	Random (mean ± sd)	0.396 ± 0.008			0.361 ± 0.086			0.333 ± 0.056		
	Head (raw)	+0.003	+0.003	+0.003	−0.090	−0.090	−0.090	+0.012	+0.012	+0.012
	All, grid	+0.004	+0.004	+0.004	+0.017	+0.017	−0.163	+0.012	−0.025	+0.012
	All, null	−0.047	+0.001	−0.003	**−0.060**	−0.064	−0.021	−0.041	+0.031	−0.009
Viewsphere v2 YOLOv5s	Random (mean ± sd)	0.453 ± 0.019			0.437 ± 0.019			0.401 ± 0.041		
	Head (raw)	+0.007	+0.007	+0.007	**+0.025**	**+0.025**	−0.017	+0.016	+0.016	+0.016
	All, grid	+0.007	+0.007	+0.007	−0.003	−0.003	−0.003	**−0.029**	−0.059	**−0.029**
	All, null	−0.081	**−0.062**	−0.021	−0.062	−0.040	−0.003	**−0.089**	**−0.089**	+0.005

As Table 8.

Table 26: Viewsphere v2 block pruning: accuracy of each ranking's cut, YOLO26

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLO26n (0.424)	Random: median (range)	0.445 (0.433–0.447)			0.411 (0.078–0.430)			0.379 (0.202–0.425)		
	Head (raw)	**0.444**	**0.444**	**0.444**	0.411	0.411	0.411	0.396	0.396	0.396
	All, grid	0.441	0.441	0.441	0.285	0.285	0.408	0.381	0.339	0.381
	All, null	0.440	0.440	0.434	**0.418**	0.406	0.416	**0.411**	0.386	0.367
Viewsphere v2 YOLO26s (0.490)	Random: median (range)	0.476 (0.465–0.487)			0.459 (0.430–0.486)			0.423 (0.365–0.454)		
	Head (raw)	0.484	0.484	0.484	**0.459**	**0.459**	0.455	**0.430**	**0.430**	**0.430**
	All, grid	0.441	0.441	0.441	0.433	0.433	0.455	0.404	0.404	0.404
	All, null	0.459	0.446	**0.485**	0.435	0.435	0.453	0.403	0.357	0.394

As Table 9.

Table 27: Viewsphere v2 block pruning: accuracy of each ranking's cut, YOLOv5

Data, model (uncut)	Ranking	$k = 1$			$k = 2$			$k = 3$		
		C	H0	H1	C	H0	H1	C	H0	H1
Viewsphere v2 YOLOv5n (0.393)	Random: median (range)	0.396 (0.384–0.407)			0.390 (0.088–0.419)			0.353 (0.150–0.411)		
	Head (raw)	0.399	0.399	0.399	0.271	0.271	0.271	0.345	0.345	0.345
	All, grid	**0.400**	**0.400**	**0.400**	**0.378**	**0.378**	0.199	0.345	0.308	0.345
	All, null	0.349	0.397	0.393	0.301	0.297	0.340	0.292	**0.364**	0.324
Viewsphere v2 YOLOv5s (0.411)	Random: median (range)	0.445 (0.425–0.475)			0.436 (0.379–0.465)			0.408 (0.290–0.447)		
	Head (raw)	0.459	0.459	0.459	**0.462**	**0.462**	0.419	**0.417**	**0.417**	**0.417**
	All, grid	**0.460**	**0.460**	**0.460**	0.433	0.433	0.433	0.372	0.342	0.372
	All, null	0.372	0.391	0.432	0.375	0.397	0.434	0.312	0.312	0.406

As Table 9.

As on v1, the raw head ranking is the one that beats random choice: YOLO26n +0.033 at $k = 3$, YOLOv5s +0.025 at $k = 2$ (both holding up), and it is never significantly worse than random. The all-block rankings mostly do not:

the null ranking is significantly worse than random in 9 of its 36 cells across the four models (down to -0.089 on YOLOv5s, $k = 3$), and significantly better in two (YOLO26n $k = 3$ with the blend, $+0.047$; YOLO26s $k = 1$ with H1, $+0.008$). Scoring on H1 alone again softens the null ranking's losses (YOLO26s $k = 1$: $+0.008$ instead of -0.018 and -0.031; YOLOv5s $k = 3$: $+0.005$ instead of -0.089). On YOLOv5n no ranking beats random by a margin that holds up.